



Figure 1: Dart logo

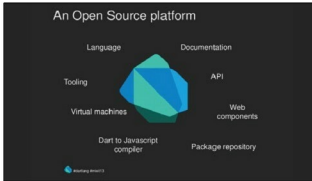


Figure 2: Dart: An open source platform

Hello, Dart!

Let's start writing a simple program in Dart by using the following commands. The syntax is very similar to any C based language.

```
//SimpleDart.dart
void main() {
  print("Hello, World!");
}
```

One of the design goals of Dart is a simple and familiar syntax. This application may be very basic, but it gives us some information about how Dart is written. First off, we can see that it uses a C-style syntax. Second, rather than JavaScript being written just about anywhere you want, our Dart code lives within a main function. That provides consistency to the code.

Although the 'Hello world' script looks simple, Dart is technologically more advanced than JavaScript and since it aims to fix security and other problems of JavaScript, it definitely has an appeal.

Now let's move on and write a more advanced Dart program.

Your first 'real' Dart program

The first thing you need to do is to set up the development environment for Dart, for which you have to install Dart Editor. It comes with the SDK and all the tools you need to get started. You can also configure any other IDE to support Dart by installing the required plugins.

We are going to keep this simple, but rather than using the command line, let's say hello to the browser. Let's create an HTML element that, when clicked, produces an alert with the

text, 'Hello, World!'

First, open Dart Editor and click on *File -> New Application* and name it *SimpleDart*. Make sure you select the command line application, and leave *Generate sample content* selected.

In your *SimpleDart* project, open the file called *pubspec.yaml* and select *Source* at the bottom of the editor. You need to add a dependency called *browser*. Here's how it looks:

```
name: SimpleDart
description: A simple Dart program.
dependencies:
  browser: any
```

Now, create an HTML file that you can use for your application's display. Right click the bin directory in your *Project navigator*, and select *New File*. Name the file *SimpleDart.html*.

Once the file is created, you should see a very basic HTML file populated with some standard elements, as shown below:

```
<p id="text"></p>
<script type="application/dart" src="SimpleDart.dart"></script>
```

The paragraph tag with the ID of the text gives you a placeholder, where you can put some text from your Dart code. Second, the script tag here is pointing to *SimpleDart*, and has a type of *application/dart*. This instructs the browser to load your Dart code and execute it from its Dart VM.

Now that you have an HTML page for your display, modify your *SimpleDart.dart* file to put some text in the paragraph tag. Before you can access the HTML though, import the appropriate library to give you HTML support, as follows:

```
//SimpleDart.dart
import 'dart:html';
```

Now create libraries that other applications can import. You are finally ready to write some code. Use a Dart method called *query*, which will search for an HTML element and allow you to modify it programmatically. Let's search for *#text*, which will give you access to the paragraph tag we saw above. Within the main function, use the following command:

```
query('#text').text = "Hey, Browser!";
```

Go ahead and run this. You will notice that the Chromium browser opens up automatically, and shows you a blank page with the text, 'Hey, Browser!' printed on it.

To add a click-event, there are a couple of options. You